Wynand van Staden

Software Engineer

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Profile

Software engineer with about two years of professional experience and an honours degree in computer science. I build software with AI coding agents: I choose the problem, direct the agent, and accept a change only once it is reproduced, tested and up to the project's standards.

Experience

Software Engineer Apr 2025 – Present
Infinity Data AI *Full-time*

- Manage data pipelines using AI coding agents.
- Contribute to and maintain a large codebase.

AI Training Contributor (part-time, freelance) Oct 2026
Superposition AI *Remote*

- Recorded narrated video-game sessions used as training data for game-playing AI models.

AI Trainer, Mathematics Nov 2024 – Mar 2025
Outlier AI *Remote*

- Wrote mathematics problems designed to expose reasoning errors in a large language model.
- Located the faulty step in the model's working and wrote the corrected solution used to retrain it.

Honours Research Project Feb 2024 – Nov 2024
Stellenbosch University *Stellenbosch, South Africa*

- Year-long research project on classifying radio galaxies by their symmetry (Python): principal component analysis for reflective symmetry, and fitted polynomials as features for curved galaxies.
- Trained in how to carry out research and report it: reading the literature, designing experiments, and writing up the method and results.

Open-Source Contributions

12 pull requests merged across 11 projects (October 2026), written with an AI coding agent under my direction. For each I picked an unclaimed bug, had it reproduced first, required a test that fails without the fix, and handled the maintainers' review. *Full list on GitHub.*

- **KAPLAY** (TypeScript, 2D game library): collisions between moving bodies ignored mass and velocity; implemented momentum exchange and tests that check momentum and energy.
- **rust-decimal** (Rust): exact parsing rejected a valid digit separator after the 28th decimal place.
- **chunkhound** (Python and Rust, code search): re-indexing an unchanged project rebuilt the whole vector index; two fixes merged.
- **smolvm**, **kloset** (Rust, Go): image extraction and backup maintenance failed on Windows.
- Also merged in **flyline**, **atlas**, **snip**, **doorstop**, **office365-rest-python-client** and **collie**.

Education

BSc (Hons) Computer Science, *Stellenbosch University, South Africa* Feb 2024 – Nov 2024
BSc Mathematics and Computer Science, *Stellenbosch University, South Africa* Feb 2020 – Nov 2023

Skills

- **AI-assisted engineering:** directing coding agents, task specification, verification-first review.
- **Languages:** Python (main); Rust, Go, TypeScript, Java and C with agent assistance.
- **Tools:** Git and GitHub pull requests, automated testing, CI, Linux and Windows, FastAPI, PostgreSQL.